Tomcat — 3–4 YOE Scenario-Based Q&A

1. Tomcat application is not starting. What will you check?
Answer:
First, I check the Tomcat logs to identify the startup or application error.
Then I verify the WAR file, Java version, configuration, port, and file permissions.
If the issue is application-related, I check the deployment logs and dependencies.
After identifying the root cause, I fix it and restart Tomcat.
Simple ga gurthupettuko:
Logs → WAR → Java → Config/Port → Fix & Restart
Commands:
tail -f $CATALINA_HOME/logs/catalina.out
ls -lh $CATALINA_HOME/webapps/
java -version
ss -lntp | grep 8080

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2. Tomcat is running but application is giving 404. What will you check?
Answer:
First, I check whether the WAR file is deployed successfully.
Then I verify the application context path and requested URL.
I also check Tomcat logs for deployment errors.
If the application is not deployed correctly, I fix the WAR or deployment configuration.
Simple ga gurthupettuko:
404 → WAR/Deployment → Context Path → URL → Logs
Commands:
ls -lh $CATALINA_HOME/webapps/
tail -f $CATALINA_HOME/logs/catalina.out
curl -I http://localhost:8080/myapp/

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3. Tomcat application is giving 500 error. What will you check?
Answer:
First, I check the application logs and Tomcat logs for the exact exception.
Then I verify application configuration, dependencies, database connectivity, and recent changes.
I identify whether the issue is from the application or backend dependency and fix the root cause.
Simple ga gurthupettuko:
500 → Logs → Exception → Config/DB → Fix
Commands:
tail -f $CATALINA_HOME/logs/catalina.out
grep -i "exception\|error" $CATALINA_HOME/logs/catalina.out
curl -I http://localhost:8080/myapp/

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4. How do you deploy a WAR file in Tomcat?
Answer:
I copy the WAR file into the Tomcat webapps directory.
Tomcat automatically detects and deploys the WAR in most standard configurations.
Then I check the logs and application URL to confirm successful deployment.
Simple ga gurthupettuko:
WAR → webapps → Auto Deploy → Logs → Verify
Commands:
cp app.war $CATALINA_HOME/webapps/
ls -lh $CATALINA_HOME/webapps/
tail -f $CATALINA_HOME/logs/catalina.out
curl -I http://localhost:8080/app/

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5. Tomcat port 8080 is already in use. What will you do?
Answer:
First, I identify which process is using port 8080.
Then I verify whether that process is required.
If it is an unwanted process, I stop it; otherwise, I change the Tomcat port in server.xml.
Simple ga gurthupettuko:
Port Busy → Find Process → Stop or Change Port
Commands:
ss -lntp | grep 8080
lsof -i :8080
grep -n "Connector" $CATALINA_HOME/conf/server.xml

6. How do you start and stop Tomcat?
Answer:
I use the Tomcat startup and shutdown scripts.
Before starting, I verify the Java environment and configuration if there was any recent change.
After starting, I check the process, port, and logs.
Simple ga gurthupettuko:
Start → Check Process/Port/Logs → Stop
Commands:
$CATALINA_HOME/bin/startup.sh
ps -ef | grep tomcat
ss -lntp | grep 8080
$CATALINA_HOME/bin/shutdown.sh

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7. Tomcat is consuming high memory. What will you check?
Answer:
First, I check the JVM heap configuration and current memory usage.
Then I check GC activity, application logs, traffic, and deployed applications.
If required, I take a thread/heap analysis to identify memory leaks or abnormal workload.
Simple ga gurthupettuko:
Memory → JVM Heap → GC → Logs → Workload/Leak
Commands:
ps -ef | grep tomcat
top
jstat -gc <pid>
jcmd <pid> VM.flags

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8. How do you check which Java version Tomcat is using?
Answer:
I check the JAVA_HOME configured for Tomcat and verify the Java version.
I also confirm that the running Tomcat process is using the expected Java installation.
Simple ga gurthupettuko:
JAVA_HOME → Java Version → Running Process
Commands:
echo $JAVA_HOME
java -version
ps -ef | grep tomcat

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9. How do you change the Tomcat HTTP port?
Answer:
I update the HTTP Connector port in server.xml.
Before restarting, I validate the configuration and check whether the new port is available.
Then I restart Tomcat and verify the new port.
Simple ga gurthupettuko:
server.xml → Connector Port → Restart → Verify
Commands:
grep -n "Connector" $CATALINA_HOME/conf/server.xml
ss -lntp | grep 8081
$CATALINA_HOME/bin/restart.sh

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10. Tomcat application cannot connect to the database. What will you check?
Answer:
First, I check the application logs for the database connection error.
Then I verify DB hostname, port, credentials, network connectivity, and database availability.
I also check whether any recent configuration or credential change caused the issue.
Simple ga gurthupettuko:
Logs → DB Host/Port → Credentials → Network → DB Status
Commands:
grep -i "database\|connection\|timeout" $CATALINA_HOME/logs/catalina.out
nc -zv <db-host> <db-port>

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11. Tomcat application is running but very slow. What will you check?
Answer:
First, I check CPU and memory utilization of the server and JVM.
Then I check GC, threads, Tomcat logs, database response time, and external API dependencies.
I identify the bottleneck before making any configuration changes.
Simple ga gurthupettuko:
CPU/Memory → JVM/GC → Threads → Logs → DB/API
Commands:
top
free -m

jstat -gcutil <pid>
ps -Lf <pid>
tail -f $CATALINA_HOME/logs/catalina.out

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12. Tomcat is running but the application is inaccessible. What will you check?
Answer:
First, I verify that Tomcat is listening on the expected port.
Then I check WAR deployment, context path, firewall/network connectivity, and application logs.
I test the application locally and then from the client side to identify where connectivity is failing.
Simple ga gurthupettuko:
Port → Deployment → Context → Network/Firewall → Logs
Commands:
ss -lntp | grep 8080
curl -I http://localhost:8080/myapp/
ls -lh $CATALINA_HOME/webapps/

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13. WAR deployment failed in Production. What will you do?
Answer:
First, I check Tomcat and application logs to identify the deployment failure.
Then I verify Java compatibility, configuration, dependencies, permissions, and the WAR file.
If the issue cannot be fixed quickly, I restore the previous working WAR and verify application health.
Simple ga gurthupettuko:
Logs → Java/Config/Dependency → Fix → Rollback
Commands:
tail -f $CATALINA_HOME/logs/catalina.out
ls -lh $CATALINA_HOME/webapps/
java -version

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14. How do you safely deploy a new WAR in Production?
Answer:
First, I test the WAR in a lower environment.
Then I take the required backup and deploy during the approved change window.
After deployment, I verify logs, application health, and critical functionality.
I keep the previous version ready for rollback.
Simple ga gurthupettuko:
Test → Backup → Deploy → Verify → Rollback Ready
Commands:
cp app.war /backup/app.war
cp newapp.war $CATALINA_HOME/webapps/app.war
tail -f $CATALINA_HOME/logs/catalina.out
curl -I http://localhost:8080/app/

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15. How do you rollback a failed Tomcat deployment?
Answer:
First, I stop the failed deployment if required and keep the previous working WAR available.
Then I restore the previous WAR and restart or redeploy the application.
Finally, I verify logs and application health.
Simple ga gurthupettuko:
Failed Deploy → Previous WAR → Restore → Restart → Verify
Commands:
cp /backup/app.war $CATALINA_HOME/webapps/app.war
$CATALINA_HOME/bin/shutdown.sh
$CATALINA_HOME/bin/startup.sh
tail -f $CATALINA_HOME/logs/catalina.out

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16. Tomcat is not starting after a configuration change. What will you check?
Answer:
First, I check the Tomcat logs for the exact startup error.
Then I validate server.xml, Java configuration, port availability, and file permissions.
If the recent configuration caused the issue, I revert the change and restart Tomcat.
Simple ga gurthupettuko:
Logs → server.xml → Java → Port → Revert
Commands:
tail -f $CATALINA_HOME/logs/catalina.out
grep -n "Connector" $CATALINA_HOME/conf/server.xml
java -version
ss -lntp | grep 8080

17. How do you check whether a WAR deployment was successful?
Answer:
I check whether the WAR is present and extracted under webapps.
Then I verify Tomcat logs for deployment success and test the application URL.
Simple ga gurthupettuko:
WAR → Extracted Folder → Logs → URL
Commands:
ls -lh $CATALINA_HOME/webapps/
tail -f $CATALINA_HOME/logs/catalina.out
curl -I http://localhost:8080/app/

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18. Tomcat logs are growing rapidly. What will you check?
Answer:
First, I identify which log file is growing and what errors are repeatedly generated.
Then I check log rotation, application logging level, and the root cause of repeated errors.
I fix the underlying issue and ensure log rotation is configured properly.
Simple ga gurthupettuko:
Growing Logs → Identify File → Repeated Error → Rotation → Fix
Commands:
du -sh $CATALINA_HOME/logs/*
tail -f $CATALINA_HOME/logs/catalina.out
grep -i "error\|exception" $CATALINA_HOME/logs/catalina.out

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19. Tomcat application is restarting repeatedly. What will you check?
Answer:
First, I check Tomcat logs and system logs for the restart reason.
Then I check JVM memory, application exceptions, configuration, and process/service status.
I also verify whether an external service or deployment process is restarting Tomcat.
Simple ga gurthupettuko:
Restart → Logs → JVM → App Error → Service/Deployment
Commands:
tail -f $CATALINA_HOME/logs/catalina.out
ps -ef | grep tomcat
systemctl status tomcat
free -m

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20. How do you troubleshoot a Tomcat Production issue?
Answer:
First, I understand the impact and check recent deployments or configuration changes.
Then I check application/Tomcat logs, CPU, memory, ports, JVM, database, and external dependencies.
I identify the root cause, apply the safest fix, and verify application health.
If the issue is deployment-related, I rollback to the last known-good version.
Simple ga gurthupettuko:
Impact → Recent Change → Logs → Resources → Dependencies → Fix/Rollback → Verify
Commands:
tail -f $CATALINA_HOME/logs/catalina.out
top
free -m
ss -lntp | grep 8080
curl -I http://localhost:8080/app/